

AFRICAN FAT-TAILED GECKO •
HEMITHECONYX CAUDICINCTUS

The calm gecko that is not a leopard gecko.

A complete, fact-checked printable manual: the floor-space enclosure, the belly heat and the humid hide this West African species needs more of than its famous cousin, exactly what to feed at every age, and the health pages that start with its most common problem, which is skin stuck on the toes.

INSIDE

- ✓ What it eats: five feeding pages, from hatchling portions to the never list
- ✓ Seven health pages, each with cause, signs, response and recovery
- ✓ Every way it differs from a leopard gecko, side by side
- ✓ Fillable owner, vet and sitter sheets, and an outage plan

42 pages

Beginner

Print or tablet

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Every page is written to be used, not just read once.

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START HERE

How to use this package

Three ways to get value from it, depending on where you are right now.

If you haven't bought a gecko yet

Read page 5 first. The African fat-tailed gecko is sold beside the leopard gecko, looks like it, and is often set up like it, and the differences on that page are the ones that decide whether a fat-tail sheds cleanly. Then read pages 6 through 10 in order before buying anything, and the feeding section after that. Use the Budget and Shopping List on page 31 to buy once instead of twice.

If your gecko is already home

Start with the Setup Checklist on page 29 to audit what you have. Then check the humid hide on page 8: retained shed on the toes and tail tip is the most common health complaint in this species, and a hide that has dried out is usually why. If you came here asking what a fat-tail eats, the feeding section starts on page 11. If anything already looks wrong, go straight to Health Red Flags on page 22.

Print these first

- | | |
|--|---|
| ☐ Page 29: Setup Checklist & Targets | ☐ Page 12: Feeding Schedule by Age |
| ☐ Page 30: Emergency & Quick Targets Card | ☐ Page 36: Pet-Sitter Sheet |
| ☐ Page 33: Symptom Quick Reference | ☐ Page 37: Owner Log |
| ☐ Page 34: Daily, Weekly & Seasonal Routine | ☐ Page 38: Equipment, Supplement & Vet Log |

How the numbers are written

Every temperature is given in °F (degrees Fahrenheit) and °C (degrees Celsius), and every length as in (inches) and cm (centimeters). Weights are in g (grams), because this gecko is small enough that a change of a few grams matters. Every abbreviation and technical word is spelled out the first time it appears.

WHY THIS PACKAGE EXISTS

The fat-tail is usually sold as a leopard gecko with a different pattern, and set up from a leopard gecko care sheet. Most of that sheet transfers. The part that does not is moisture: this gecko comes from the savannas of West Africa rather than the dry rock country of southern Asia, it needs a wetter enclosure and a humid hide that is never allowed to dry, and the most common fat-tail health problem traces straight back to that gap. Every figure here matches the Beastly Facts fat-tail guides or is cross-checked against veterinary sources, listed on page 41.

NOT A SUBSTITUTE FOR

A veterinarian who sees reptiles. This is a husbandry reference for day-to-day care, and anything urgent is a professional call, not a page in this book. Not every practice treats reptiles, so find one before you need one and write the number on page 30.

SECTION 01

Quick profile

Everything at a glance before you read the detail.

Scientific name	<i>Hemitheconyx caudicinctus</i> . A eublepharid, one of the eyelid geckos: the same family as the leopard gecko, with true eyelids and no sticky toe pads
Native range	West Africa, from Senegal to Cameroon. Savanna, dry forest and rocky scrubland, sleeping by day under rocks, surface debris or in an abandoned burrow
Adult size	7 to 9 in (18 to 23 cm). Hatchlings are about 2 in (5 cm). Males grow larger than females
Adult weight	40 to 90 g, averaging about 50 g. Page 17
Captive lifespan	15 to 20 years is the commonly cited range, and the best-documented captive record is just over 16 years. Plan on well over a decade
Activity	Nocturnal. Asleep in a hide by day, hunting on the ground at night
Diet	Live insects only: an obligate insectivore, meaning it must eat insects and cannot use plant food. Page 11
Enclosure, one adult	36×18×18 in (91×46×46 cm) as the minimum. A 40-gallon breeder tank, 36×18×16 in (91×46×41 cm), gives the same floor. Floor space, not height
Temperature	Warm side 88 to 92°F (31 to 33°C) at the surface, cool side 75 to 80°F (24 to 27°C), night 70 to 75°F (21 to 24°C). Page 7
Humidity	50 to 70% through the enclosure, with a permanent humid hide at 70 to 80%. Higher than a leopard gecko. Page 8
UVB	UVB (ultraviolet B) is the band of ultraviolet light a reptile uses to make vitamin D3 in its skin. Optional for this species if it gets a calcium powder with D3, but a low-output 2 to 5% UVB tube is increasingly recommended. Page 10
Most common problem	Retained shed on the toes and tail tip, almost always from low humidity or no humid hide. Page 23
Handling	One of the calmest pet geckos, often called more docile than the leopard gecko. Never by the tail
Conservation status	Least Concern on the IUCN Red List, the global list of species at risk kept by the International Union for Conservation of Nature. Buy captive-bred

Cost overview

Initial setup

The gecko is \$75 to \$600, with normals commonly around \$100 and fancier color lines, called morphs, costing more. Equipment comes to **\$369 to \$579**. With a first exam and fecal test (a stool check for parasites), plan on **\$469 to \$794** before the gecko. Page 31.

Ongoing

Feeders and supplements run **\$10 to \$25 a month**, or \$120 to \$300 a year. With a yearly exam and fecal test, a healthy adult runs about **\$220 to \$515 a year**. A sick visit with tests is typically \$150 to \$300 or more.

THE NUMBER THAT SHOULD DRIVE THE DECISION

Not the price. The lifespan. Over 15 years, the yearly costs alone come to about \$3,300 to \$7,700 on top of the gecko and the first setup. The purchase is the cheap part, and the routine on page 34 is what you are actually signing up for.

Fat-tail or leopard gecko? The differences

Same family, same shape, same shop shelf. Not the same care sheet.

The African fat-tailed gecko and the leopard gecko are both eublepharids: ground-dwelling geckos with eyelids, no climbing pads, a fat-storing tail, and a habit of dropping that tail when grabbed. Most of what a leopard gecko owner knows does transfer. Three things do not: humidity, temperament and origin. A fat-tail kept to a leopard gecko sheet usually goes wrong on humidity.

	AFRICAN FAT-TAILED GECKO	LEOPARD GECKO
Origin	West Africa, Senegal to Cameroon: savanna, dry forest and rocky scrub	Pakistan, Afghanistan, Iran, India and Nepal: rocky desert, arid grassland and dry scrub
Humidity	50 to 70% through the enclosure, plus a humid hide at 70 to 80% that is a permanent fixture	30 to 40% through the enclosure, with a humid hide at 70 to 80%
Temperament	Among the calmest pet geckos, often described as even more docile. More vocal than most pet geckos	Calm and easy to handle once settled
Underfoot	A burrower: a soil and sand mix at least 4 in (10 cm) deep for healthy adults	Paper towel or tile is the usual default, with a dig box as enrichment
Belly heat	88 to 92°F (31 to 33°C) at the warm side surface, cool side 75 to 80°F (24 to 27°C)	88 to 92°F (31 to 33°C) at the warm hide, cool side about 70 to 77°F (21 to 25°C)
Enclosure	36×18×18 in (91×46×46 cm) minimum, past the 20-gallon long that veterinary care sheets still print	36×18×18 in (91×46×46 cm) is now the minimum
Nest choice	Gravid females, those carrying eggs, picked nest sites averaging 32.4°C (about 90°F) in a temperature-choice study	The same study: 28.7°C (about 84°F)
Price	\$75 to \$600, normals around \$100	\$20 to \$45 for a standard animal from a store

Why humidity is the one that matters

Stuck shed on the toes and the tip of the tail is the most common fat-tail-specific health complaint, and it shows up when the enclosure runs too dry or the humid hide is missing or dried out. A ring of old skin left on a toe tightens as it dries and can cut off the blood supply. A leopard gecko sheet, followed exactly, produces the drier enclosure that causes it. The fix is not a wet tank: the enclosure stays mostly dry, at 50 to 70%, and the moisture lives in one damp hide. Page 8 has the setup and page 23 the problem.

DO NOT RAISE HUMIDITY EVERYWHERE

Going the other way is its own mistake. An enclosure that is too damp and too cool at the same time is what causes respiratory infection in this species (page 25). Humid in the hide, moderate in the air, warm at one end.

SAME FAMILY, SAME HABITS

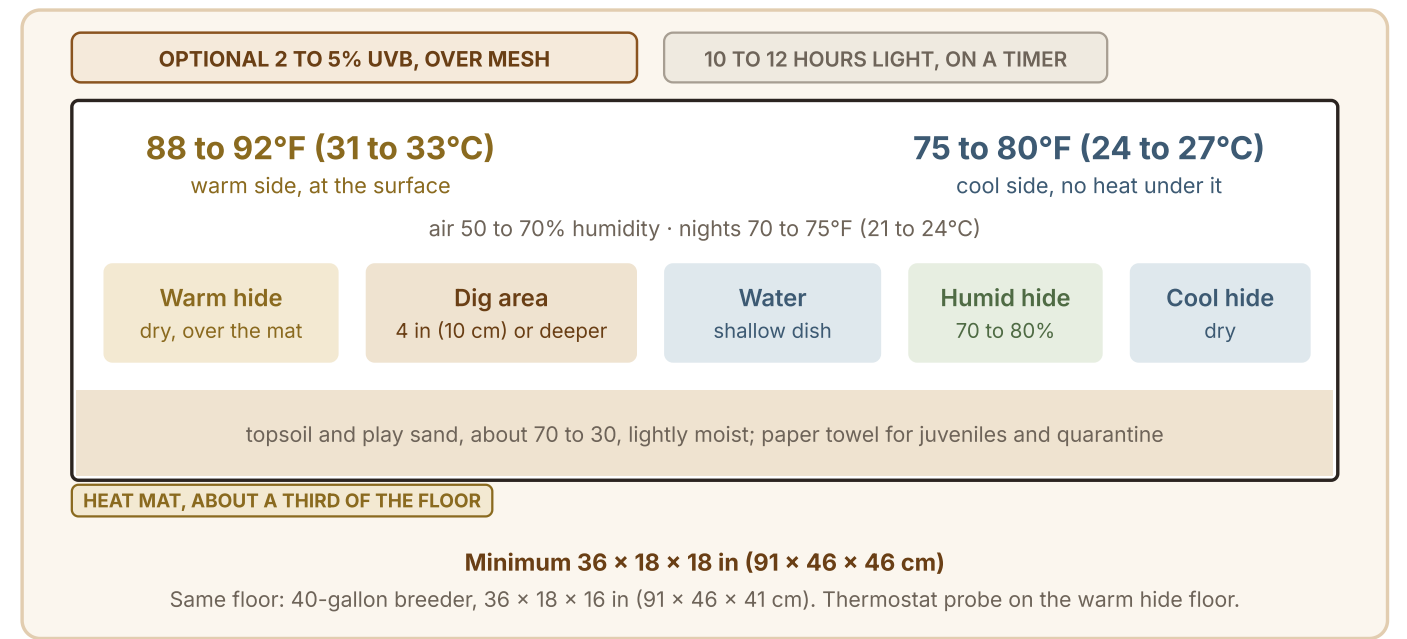
What carries straight across from a leopard gecko: belly heat from a mat on a thermostat, three hides, live insects dusted with calcium, a tail that is never a handhold, and the tail as the body-condition gauge. If you have kept a leopard gecko, you already know most of this book. Read the humidity and substrate pages twice.

A floor-space enclosure & where it goes

A ground-dweller and a burrower. It lives on the floor and under it, so buy floor.

36×18×18 in (91×46×46 cm) is the minimum for one adult, and a **40-gallon breeder, 36×18×16 in (91×46×41 cm), gives the same floor**. The 20-gallon long, about 30×12×12 in (76×30×30 cm), that many care sheets still print is the old floor. Height buys almost nothing for a terrestrial species. The extra length is what lets one end run warm and the other cool, and the extra depth is room for the deep digging substrate on page 9. Skip the tall branches and climbing walls that arboreal geckos need.

One gecko per enclosure. Both sexes are territorial, males fight over space and females, and two males must never share. A second fat-tail means a second full setup. A single gecko loses nothing: in the wild this species is solitary and rarely seen in groups.



Where the enclosure goes

PLACE	WHY
Out of direct sun	Sun through glass heats a closed box far past the warm-side target, and the gecko cannot leave. Heat comes from the thermostat-controlled mat, never the window
Off radiators, vents and drafts	Each one swings the temperature and dries the air, which fights the humidity on page 8
Not in the kitchen	Reptiles carry <i>Salmonella</i> bacteria with no sign of illness, and the hygiene rules on page 16 start with keeping the gecko and its gear away from food
Where you will see it	This gecko sleeps through the day and comes out at dusk. Put it where you will be in the evening, because that is when the daily health check on page 34 happens

A front-opening enclosure, or a tank with a secure mesh lid, both work. The mesh lid matters because glass blocks UVB (page 10).

Belly heat, the thermostat & the probe

Unlike some geckos, this one needs supplemental heat, and it takes that heat through its belly.

ZONE	°F	°C
Warm side, at the surface the gecko lies on	88 to 92	31 to 33
Cool side	75 to 80	24 to 27
Night	70 to 75	21 to 24
Act if the night falls below	70	21

The heat source

An **under-tank heat mat covering about a third of the floor**, always on a thermostat, with the warm hide sitting directly over it. A mat under a thick layer of substrate heats the surface poorly, so put the warm hide on the mat and keep the deep digging area off it. **Skip heat rocks and strong overhead basking bulbs entirely**: both carry real burn risk for an animal that presses its belly to the warmest spot it can find.

The thermostat is the safety device

A heat mat has no idea how hot it is. Left unregulated it climbs past the target on a warm afternoon, and a gecko seeking belly heat stays on the hot spot rather than moving off. That is how thermal burns happen. A basic **on/off thermostat** suits a mat, since a mat changes temperature slowly enough that a full on and off cycle stays gentle. A faster heat source such as a ceramic heat emitter, a screw-in heater that gives heat and no light, is better on a **pulse proportional or dimming** unit, which eases power down before it overshoots.

Where the probe goes

The probe goes **flat on the floor of the warm hide**, the surface the gecko actually lies on, held down with tape so it cannot drift. Never taped to the back of the mat, which reads the mat's own temperature, and never hanging in the air. A probe that slips off its spot fails silently: the display keeps showing a correct number for the wrong place. Check its position at every water change.

Measure it two ways

- ☐ An infrared temperature gun, which reads a surface from a distance, aimed at the warm hide floor: the number that matters
- ☐ A second digital thermometer, apart from the thermostat probe, read daily. A thermostat stuck on, or a probe the gecko has shifted, gives no other warning
- ☐ A digital probe thermometer at the cool end. One reading tells you nothing about a gradient

COLD NIGHTS

The mat keeps working at night with the lights off. If the room itself takes the enclosure below 70°F (21°C), add a heat source that gives no light, such as a ceramic heat emitter on its own thermostat, set well clear of anything the gecko can reach. A gecko kept too cool can stop eating as if a winter slowdown had started, and a cool, damp enclosure is the classic setup for respiratory infection.

Humidity & the three hides

Higher than a leopard gecko needs, but this is still a mostly dry enclosure. The moisture lives in one box.

WHERE	TARGET
The enclosure air, read mid-height	50 to 70%
Inside the humid hide	70 to 80% , permanently

The humid hide is not optional here

A fat-tail needs a persistently damp retreat as a standing feature, not a shedding aid set out now and then. A closed gecko cave packed with **damp sphagnum moss**, or damp paper towel, stays humid inside while the rest of the enclosure runs drier. Damp means it holds together when squeezed and does not drip. This one hide is what prevents almost all of the stuck shed on page 23, and it helps keep the gecko hydrated. A humid hide that has dried out is only a hide, so check it every day.

Three hides, minimum

WARM HIDE

Dry, snug, directly over the heat mat, with the thermostat probe on its floor. Where the gecko digests.

HUMID HIDE

Damp moss inside, at the cool end or the middle. Directly over the mat it bakes dry in hours.

COOL HIDE

Dry, snug, at the far end. Where the gecko goes to cool down without giving up cover.

The point of three is that the gecko can be hidden **and** at the temperature and moisture it wants, at the same time. One hide forces it to choose between feeling safe and being comfortable, and it will choose safe.

Holding the numbers

- **Read a digital hygrometer** (a humidity meter) mid-enclosure, away from the humid hide, a few times a week, and at the same hour each time.
- **Too dry in a dry house?** A light misting a few times a week raises the air for a while. A lightly moist lower layer of substrate helps too. Let the hygrometer decide, not the calendar.
- **Too wet?** Condensation that sits on the glass, a damp or sour smell, or mold. Mist less and let air move. A damp enclosure that is also too cool is the respiratory infection setup.
- **Refresh the moss** whenever it smells sour or looks soiled. A warm, damp hide is where waste and mold build up fastest.

NEVER

Raise the whole enclosure to 70 to 80% to make up for a missing humid hide. That is the leopard gecko mistake in reverse, and it trades stuck shed for a respiratory risk. Some keepers run a seasonal humidity cycle; a humid hide that never dries matters more than chasing an exact seasonal number.

Substrate for a burrower & furnishings

The detail that makes this species different from most pet geckos: it digs.

WHO	SUBSTRATE
Healthy adults	A naturalistic sandy-soil mix, at least 4 in (10 cm) deep so the gecko can burrow. A common blend is roughly 70% topsoil to 30% play sand , or coconut fiber mixed with reptile sand. Use untreated topsoil, with no fertilizer or added chemicals. Keep it lightly moist rather than dust-dry, which also helps hold a burrow
Juveniles	Paper towel
Any gecko in quarantine	Paper towel, for the whole quarantine. It shows stool, mites (tiny parasites that live on the skin) and a regurgitated meal that loose substrate hides (page 19)
An unwell gecko	Paper towel until it is well again

The sand question, settled

Never keep one on **pure sand**: it dries the gecko out and interferes with shedding, which is reason enough on its own. Avoid loose calcium sand too. The impaction half of the sand argument, the risk of a gut blockage, is narrower than it sounds. Sand swallowed with prey tends to cause trouble in a gecko that is already unhealthy, dehydrated or kept too cool, not in a healthy one on a soil mix. That is why juveniles, unwell animals and quarantine geckos go on paper, and why prey is sized and offered from tongs (page 12).

Furnishings checklist

- ☐ Warm dry hide, snug, over the mat
- ☐ Cool dry hide at the far end
- ☐ Humid hide with damp sphagnum moss
- ☐ Shallow water dish, refreshed daily
- ☐ Cork flats and stacked slate, low and stable
- ☐ A deep patch of diggable substrate off the mat
- ☐ For any female: a laying box (page 18)
- ☐ Low branches only if wanted; height is secondary

Every piece of decoration has to be stable and able to take the gecko's full weight. A burrower digs under things, so set slate and rock on the tank floor before the substrate goes in, not on top of it.

Keeping it clean

HOW OFTEN	WHAT
As you see it	Remove droppings and uneaten insects
Daily	Empty, rinse and refill the water dish. Check the humid hide is damp
When sour or soiled	Replace the moss in the humid hide
Weekly	Wipe down hides, dish and glass
Monthly	Deep clean, and replace the substrate. Household bleach at a 1:32 dilution, which is half a cup per gallon of water, kept wet on the scrubbed surface for 10 minutes, then rinsed until no smell remains and dried

Light cycle & UVB (ultraviolet B) light

A nocturnal gecko still needs a day. Whether it needs UVB is a choice with one rule attached.

The day length

Run lights for **10 to 12 hours a day** on a plug-in timer, so the cycle never depends on someone remembering. At night the lights go off. This gecko is awake in the dark, and the heat mat keeps the warm side warm with no light at all.

UVB: optional, and increasingly recommended

UVB is the part of ultraviolet light that lets a reptile make vitamin D3 in its own skin, and vitamin D3 is what lets it use the calcium it eats. Fat-tailed geckos are nocturnal and can live without UVB **if they get vitamin D3 from a calcium powder with D3** instead. Low-output UVB in the **2 to 5%** range is increasingly recommended anyway. Either path is acceptable. No path is acceptable that gives the gecko neither, because that is how metabolic bone disease starts (page 24).

IF YOU RUN UVB	HOW
Strength	A low-output tube rated 2 to 5%. Researchers sort reptiles into four Ferguson Zones by the ultraviolet exposure they seek in the wild, and this gecko sits in Zone 1, the shade dwellers and the crepuscular species, those most active at dusk and dawn. So the target UVI (ultraviolet index, the measured strength of UVB where the gecko sits) is 0 to 0.7 across the enclosure and 0.6 to 1.4 at the closest point
Placement	Along the warm side, inside the enclosure or over a mesh top. Never over glass , which blocks most UVB. A mesh lid costs some output; a glass lid costs most of it
Shade	The hides are the shade. A gecko that wants no UVB goes under cover, which is exactly how a shade-dwelling species uses it
Replacement	About every 12 months, long before the tube stops looking bright. UVB output fades well before visible light does. Some models need it sooner; follow the maker's interval
Timer	On the same timer as the day cycle

If you do not run UVB

Then the calcium with D3 days on page 14 are not optional, and the dusting tally on page 38 is worth keeping. A gecko without UVB depends on the powder for all of its vitamin D3.

DO NOT STACK BOTH AND THEN ADD MORE

Vitamin D3 is fat-soluble and builds up in the body, so more is not safer. A gecko under a working UVB tube is already making its own; keep its calcium with D3 at the low end of the schedule, and never dust D3 at every feeding.

WHAT A FAT-TAIL DOES WITH LIGHT

It sleeps through most of it. Expect to see your gecko at dusk and after the lights go out, not under the lamp. A gecko that is out and moving in its evening window is a healthy sign in itself (page 21).

What African fat-tailed geckos eat

The short answer, then the four pages that make it work.

THE SHORT ANSWER

Live insects, and nothing else. An African fat-tailed gecko is an obligate insectivore: it must eat insects and cannot digest fruit, vegetables or plant-based gecko diets. The staple feeders are **crickets, dubia roaches, discoid roaches, mealworms, superworms and black soldier fly larvae**, captive-bred and gut-loaded (fed a calcium-rich diet before they are offered), dusted with plain calcium at most feedings, and offered in the evening. Fresh water in a shallow dish, always.

The four parts of a fat-tail diet

PART	WHAT IT IS	PAGE
The insect	Six staples, four occasional higher-fat treats, and a list of insects that must never be fed. Captive-bred only, never wild-caught	13
What the insect ate	Gut-loading: feeding the insects a calcium-rich diet for 24 to 48 hours first, because the gecko eats whatever is in their gut	14
The powder	Plain, phosphorus-free calcium at most feedings, calcium with vitamin D3 once or twice a week, a multivitamin about once a week	
How much, how often	Daily for hatchlings, easing to 3 times a week or less for adults. About 2 insects per inch of the gecko's length, none wider than the space between its eyes	12

A sample week for an adult

One way to lay out an adult fed 3 times a week. Every feeding gets calcium; vitamin D3 comes once or twice; the multivitamin once.

EVENING	INSECTS	DUSTING
Monday	Gut-loaded crickets	Plain calcium, plus the weekly multivitamin
Wednesday	Gut-loaded dubia roaches	Calcium with D3
Friday	Crickets, or black soldier fly larvae	Plain calcium
Other evenings	Nothing	Fresh water, a look at the tail

If your multivitamin already contains D3, that day counts as one of the D3 days. Without UVB, a second calcium with D3 day is the usual choice. With UVB, one is enough.

Three things people get wrong

- **Feeding fruit, baby food or crested gecko powder.** Not food for this animal, whatever a video shows.
- **Catching insects in the yard.** Pesticides and parasites, and some common insects are toxic outright.
- **Counting days when it stops eating.** A healthy adult can fast for weeks on its tail. Watch the tail, not the calendar (page 15).

SECTION 03 • FEEDING

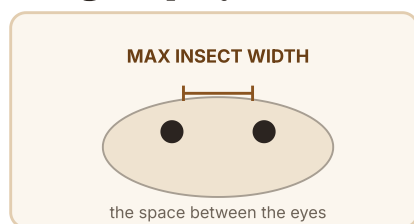
Feeding schedule by age & how much

Daily while it grows, then less often. The tail tells you whether the schedule is right.

AGE	HOW OFTEN	HOW MUCH
Hatchling , under about 4 months	Daily	About 5 crickets, each around 1/4 in (6 mm) long, sizing up as the gecko grows
Juvenile	Daily for a still-growing animal, easing toward 3 times a week	About 9 crickets or mealworms a feeding at the 3-times-a-week end. Move toward the adult schedule as the tail fills out
Young adult	Every 2 to 3 days	About 2 insects per inch of body length, or what it eats in about 15 minutes
Adult	3 times a week, or 3 to 4 times	As above
Mature adult with a visibly fat tail	Every 5 days	As above

The sources give these adult intervals as a range rather than one number, so treat them as a range and let the tail decide. Fat, as thick as the neck or thicker, and holding steady: the schedule is right. Thinning: feed more often, and see page 15 if it is refusing.

Sizing the prey



No insect wider than the space between the gecko's eyes. Oversized prey is an impaction risk, and so is a mouthful of substrate taken with a cricket that ran into it. Measure against the gecko, not the label on the cup: hatchlings take insects about 1/4 in (6 mm) long, and the size goes up with the gecko.

How to offer it

- **In the evening.** A nocturnal gecko responds to food offered at dusk far better than to food offered while it should be asleep.
- **From tongs, or in a smooth-sided dish.** Feeding tongs control the portion and keep insects from burrowing into the substrate, which is where they get lost and where substrate gets swallowed.
- **Remove what is not eaten** after about 15 minutes, or by the next morning at the latest.
- **Fresh water in a shallow dish at all times**, changed daily.
- **An elderly gecko** can lose teeth and struggle with hard-bodied prey such as crickets. Softer prey keeps it eating.

A TAIL-RAISE IS A GOOD SIGN

A fat-tail stalking prey often lifts its tail and waves it slowly. That is hunting behavior, not distress, and it means the feeding is doing its job. Varying where and when prey appears keeps it that way (page 20).

Feeder insects: staples, treats & never

Captive-bred only, gut-loaded first, and no wider than the space between the eyes.

INSECT	TIER	NOTES
Crickets	STAPLE	The usual staple, cheap and active. On their own they run about 1 part calcium to 9 parts phosphorus, which is why they are gut-loaded and dusted
Dubia roaches	STAPLE	Somewhat better on calcium than crickets, still short of it. Easy to gut-load
Discoid roaches	STAPLE	The same role as dubia
Mealworms	STAPLE	A staple in rotation rather than the whole diet: the worst calcium balance of the common feeders, about 1 to 14
Superworms	STAPLE	Large, so for adults only, and sized against the eye rule like everything else
Black soldier fly larvae	STAPLE	The one common feeder that is not calcium-poor: it carries calcium in its skin and lands near 1 to 1 unaided. Dust it anyway
Silkworms	OCCASIONAL	An occasional, higher-fat supplement, not a staple
Hornworms, commercially raised	OCCASIONAL	Occasional only, and only from a feeder supplier. Wild hornworms from tomato plants are on the never list
Waxworms	SPARINGLY	High in fat. A treat, sparingly
Pinkie mice	RARELY	Listed as an occasional item; rarely, as a treat, for an adult

The never-feed list

NEVER	WHY
Fireflies, ladybugs, monarch butterflies, box elder bugs, assassin bugs, lubber grasshoppers	Toxins, or a wild-caught contamination risk, or both
Wild hornworms taken from tomato plants	Wild-caught: a toxin and contamination risk
Centipedes, hairy or spiny caterpillars, bees and wasps	Toxin and contamination risk, and bees and wasps sting
Any insect caught outdoors, however clean the yard looks	Pesticide and parasite exposure
Fruit, vegetables, baby food, fruit-based gecko powders	An insectivore's gut cannot use them
Anything wider than the space between the eyes	An impaction risk (page 25)

ROTATE THE STAPLES

The staples differ in fat and in calcium balance, and no single one is the best at everything. Two or three in rotation, gut-loaded and dusted, cover more ground than any one alone. The higher-fat treats stay treats, because the tail stores what the gecko does not burn.

Gut-loading & calcium dusting

A gecko's nutrition is mostly decided before the insect is ever offered.

Why insects need help

A gecko needs more calcium than phosphorus. The accepted target for insect-eating reptiles is roughly **2 parts calcium to 1 part phosphorus**, because phosphorus competes with calcium for absorption. Feeder insects come nowhere close: crickets sit near 1 to 9, mealworms near 1 to 14. Two habits close the gap. **Gut-loading** fills the insect with calcium-rich food before it is eaten, and **dusting** coats it in calcium powder. Do both: you cannot see how much a given cricket actually ate.

Gut-loading

Feed the insects **24 to 48 hours before offering them**, inside the 24 to 72 hour window where gut-loading works. Gut-loading is not cumulative: an insect carries only what is in its gut right now, so offer insects soon after they come off the food.

FEED THE INSECTS	VERDICT	WHY
A commercial high-calcium gut-load diet	USE	The reliable option. Buy one sold on its calcium content
Collard, mustard, turnip or dandelion greens	USE	Calcium-dense and low in oxalates. Dandelion only from ground that has never been sprayed
Butternut squash	USE	Calcium-dense, and insects take it readily
Calcium-fortified water gel	USE	Insects drown in open water. A gel gives water and calcium by the same route
Carrot, potato, apple	NOT ENOUGH	Keeps insects alive, does not move the calcium balance
Spinach, beet greens, Swiss chard	NEVER	High in oxalic acid, which binds calcium so nothing can absorb it

The dusting schedule

POWDER	HOW OFTEN	NOTES
Plain calcium , no D3, phosphorus-free	At most feedings: every feeding at the strict end, at least 3 times a week	The everyday powder. Read the label: skip any calcium that also contains phosphorus , which can damage the kidneys and cause gout, a painful buildup of uric acid in the joints and organs
Calcium with D3	Once or twice a week, on top	The vitamin D3 source for a gecko without UVB. Use the low end under a working UVB tube
Reptile multivitamin	About once a week	Covers vitamin A and the rest. If it contains D3, count that day as a D3 day

To dust, put the insects in a cup or bag with a pinch of powder and shake gently until they look lightly floured, then offer them straight away.

NEVER DUST D3 AT EVERY FEEDING

Vitamin D3 is fat-soluble and builds up in the body, so more is not safer. Plain calcium is the routine; calcium with D3 is once or twice a week. Write your dusting days on page 30 so a sitter does not double them.

Why it stops eating & when to worry

This gecko can go months without eating and be fine. That makes the tail the question, not the day count.

The common reasons

1. **The season.** As days shorten in fall and winter, adults commonly eat less or stop for a while. Juveniles usually keep eating through it.
2. **Breeding season.** Females may go off food for a few days before laying, within a breeding season of roughly 4 to 5 months (page 18).
3. **A shed.** Shedding takes energy and often comes with a short appetite drop. If the shed sticks, including retained eye caps, the gecko can turn lethargic and stop eating until it is dealt with (page 23).
4. **Stress.** A new home, or recent handling. Leave it alone until it is eating and passing stool reliably.
5. **An enclosure that is too cool.** A cold gecko can act as if a winter slowdown has started, out of season. A husbandry fix, not an emergency by itself: measure the warm hide floor first.
6. **Dehydration.** Too little humidity or water access suppresses appetite.
7. **Illness.** Internal parasites, bacterial infection or, rarely, tumors, usually with weight loss and lethargy.
8. **Old age.** Lost teeth make hard-bodied prey difficult. Softer prey helps.
9. **A long picky phase.** Otherwise healthy geckos have been recorded refusing food for months while holding a stable weight, the tail buffering the gap.

What to check first

- | | |
|---|---|
| ☐ Warm hide floor with the infrared gun: 88 to 92°F (31 to 33°C) | ☐ Tail: as thick as the neck, or thinner? |
| ☐ Humid hide still damp, and no skin stuck on toes, tail tip or eyes | ☐ A female digging or restless: is she about to lay? |
| ☐ Weight in grams against the log on page 37 | ☐ Stool and urates, the white part of a dropping, in the last week (page 28) |

WHEN A REFUSAL BECOMES A VET VISIT

There is no universal day count. What matters is real weight loss, as opposed to a tail that is a little less plump: **a tail thinner than the neck is the warning sign.** Go to the vet for weight loss, a bloated belly, lethargy or weakness beyond a normal seasonal slowdown, retained shed that does not clear with humidity and soaking, or signs of dehydration alongside not eating. Shedding-related dehydration that runs past about a week can become fatal.

JUVENILES HAVE LESS MARGIN

Juveniles usually keep eating through the fall and winter drop, so a juvenile that stops is not taking a seasonal break. It gets the checks above the same day, and a vet call if it does not resolve.

Handling a calm gecko

Among the calmest, most handleable pet geckos there are. The rules are few and they matter.

Adults tolerate handling and usually stay calm through a session. Juveniles tend to be shy and take longer to settle. Any gecko can bite if it is sick, scared or provoked, but for this species a bite is the exception. The calm has a context: in the wild this is a nocturnal savanna gecko that spends the day asleep under cover, so handling goes best in the evening, when it is already awake and moving.

Getting started

- **Wait about two weeks** after a new gecko comes home, and do not start until it is eating regularly.
- **Approach from the side.** A hand that comes in from the side is easier on it than one that scoops from above.
- **Support the whole body** and let it walk across your hands rather than gripping it. Keep early sessions short.
- **Stay low**, over a bed, couch or the floor. A gecko that walks off a hand at chest height is a fall risk.
- **Not right after a meal, and not during a shed.**

Reading it in the hand

This species is more vocal than most pet geckos, and the sounds mean something. In the hand, **a chirp or a squeak signals discomfort**: the cue to ease off and put it down. A slow tail-swish combined with an arched back is a clearer "leave me alone". A slow tail-wave while stalking an insect is something else: hunting.

NEVER BY THE TAIL

Like most geckos, a fat-tail drops its tail if grabbed roughly or badly frightened. This is called autotomy. The tail regrows within weeks, but comes back smoother and more bulbous, and the fat reserve it carried is gone. Never grab or restrain the tail, and tell every visitor the same.

Handling as a health check

Each session is the time to look the gecko over: a plump tail, no old skin on the toes or tail tip, clear eyes. Weigh it on a kitchen scale that reads in single grams and log the number once a month (page 37).

SALMONELLA

Healthy reptiles carry *Salmonella* bacteria in their droppings without looking ill. A CDC (United States Centers for Disease Control and Prevention) investigation into infections linked to pet geckos, closed in January 2026, counted 113 cases and 31 hospitalizations across 36 states. Wash your hands with soap after touching the gecko, its food or anything in its enclosure. Keep it and its gear out of the kitchen, and never clean gear in a kitchen sink or a bathtub people use: keep a plastic tub for it. Do not kiss the gecko. Children under 5 should not handle reptiles at all, and adults over 65, pregnant people and anyone with a weakened immune system are at higher risk.

Sexing, weight & body condition

Telling male from female

Sexing is done by looking at the underside near the vent, the single rear opening that waste and eggs pass through, and it only works on maturing animals. **Females mature at roughly 250 to 300 days after hatching, males at 300 to 350 days**, and males grow larger than females from around then. A mature male shows:

- A chevron-shaped row of enlarged **preanal pores** just in front of the vent: small openings that produce a waxy scent secretion.
- A proportionately more bulbous tail base behind the vent, where the **hemipenes**, the paired male reproductive organs, sit.

Females lack the enlarged pore row and have a flatter tail base. Never hold the gecko by the tail to look.

Hatchlings: the incubation clue

Sex in this species is set by the temperature the egg was incubated at, which is called temperature-dependent sex determination. A breeder who records it can tell you which way a clutch was likely to go:

INCUBATED AT	MOSTLY
24 to 28°C (75 to 82°F)	Female
29 to 30.5°C (84 to 87°F)	A roughly even split
31 to 32°C (88 to 90°F)	Male

A temperature is a probability. A young gecko sold as "probable female" gets checked again once the pores appear, and it matters: two males must never share an enclosure.

Weight

Adults weigh 40 to 90 g, averaging about 50 g. No veterinary growth table by age has been published for this species, so the useful number is your own gecko's trend. A loss of 5 g is a tenth of an average adult and invisible by eye until the tail has already thinned. Weigh monthly, more often if anything looks off, and log it.

Body condition at a glance

TOO THIN

A tail thinner than the neck. A tail that thins to skin over bone, sometimes called stick tail, means underweight and a vet check for parasites. Rule out a cool warm side first.

RIGHT

A plump tail, as thick as the neck or thicker; in a well-fed adult it can run about 1.25 in (3 cm) thick. Weight steady in an adult, rising in a juvenile.

TOO HEAVY

Obesity in reptiles comes from too many calories and too little activity. Bring it down slowly over months, with the vet, by cutting fatty treats and spacing feedings. Not a crash diet.

Females, eggs & the laying box

A female kept entirely alone can still produce eggs. This page is for every female owner, not just breeders.

Eggs without a male

Female egg-laying reptiles can form eggs with no male present, so every adult female is at risk of egg binding, a female unable to pass the eggs she has formed. Those eggs are infertile and will not hatch, but the physical work of forming and laying them is the same. Females can also store sperm from an earlier mating, so a female who once lived with a male can lay fertile eggs later.

The laying season

WHAT	TYPICAL
Breeding season	Roughly 4 to 5 months a year
Eggs per clutch	1 to 3
Clutches	Up to 5 in a season, many females fewer, often 15 to 22 days apart
Appetite	May drop for a few days before laying
If fertile: incubation	6 to 12 weeks, shorter when warmer

Signs she is about to lay

- Restlessness and repeated digging, in corners, in the substrate or in the humid hide.
- Going off food for a few days.

Give her somewhere to lay

A missing or unsuitable nesting site is one of the recognized causes of egg binding, and it is the easiest to remove. Every adult female gets a **laying box**: a covered box of moist substrate, such as damp sphagnum moss or a damp soil mix, deep enough for her to dig into, set over the warm end. Fat-tail females are particular about temperature when they lay: in a gradient study they chose nest sites averaging 32.4°C (about 90°F), warmer than leopard geckos chose, so the box belongs where the warm side reaches. Leave her alone with it.

AFTER SHE LAYS

Remove infertile eggs, which will not hatch. Offer food and water. Weigh her, so the next clutch is readable against this one. Poor nutrition is among the causes of egg binding, so a laying female's calcium dusting is not the place to cut corners.

EGG BINDING IS A SAME-DAY VET CALL

Straining, a swollen vent, restlessness that does not settle, repeated digging with nothing produced, or a gravid female, one carrying eggs, who stops eating and turns dull. Page 26 has the full entry.

Choosing a gecko, quarantine & the law

The healthiest gecko is chosen well, kept apart until it has proved itself, and legal where you live.

Captive-bred, not wild-caught

For years most fat-tails offered for sale were wild-caught adults, exported from West Africa through Ghana, Togo and Benin, and wild-caught animals still turn up. Captive breeding has since caught up. **Buy captive-bred.** A parasite load matters most in a wild-caught gecko, and veterinary care sheets single out new and wild-caught geckos for fecal exams. Ask the seller four things: the hatch date, whether it was captive-bred, what it eats and when it last ate, and whether it has been treated for anything. A good breeder answers all four without looking anything up.

Pass on any gecko showing a tail thinner than its neck, sunken eyes, skin stuck on the toes or tail tip, difficulty breathing, diarrhea, an injury, or real lethargy in the evening when it should be awake.

Quarantine: 3 to 6 months

If the new gecko will ever share a room, the air or your hands with another reptile, quarantine it. Veterinary husbandry guidance gives **3 to 6 months** for a new reptile; the 60 to 90 days often repeated in the hobby is the floor, not the target.

- ☐ A separate room, not a second tank on the same shelf
- ☐ Paper towel substrate, which shows mites, stool and a regurgitated meal
- ☐ Its own tongs, dishes and scraper, never shared
- ☐ Established animals cared for first, the new one last, hands washed between
- ☐ A vet exam with a fecal test inside the window
- ☐ Disinfect everything it touched before reuse

A clear fecal result is a good sign, not the end of quarantine: the window runs its full length. If this is and will remain your only reptile, the collection argument does not apply, but an early vet visit with a fecal test is still worth it.

NEVER RELEASE A GECKO OUTDOORS

Not a healthy one, not an unwanted one. A gecko you cannot keep goes back to the breeder, to a rescue, or to a reptile society.

Where it is legal

The African fat-tailed gecko is legal to keep in most of the United States, and no state-by-state review of this species has been done for this package, so it carries no state table. Some rules catch every lizard or every gecko regardless of species: the **District of Columbia's** list of allowed animals includes no lizard; **Hawaii** allows an animal only if it appears on its approved lists, so anything never written down is barred; **West Virginia** treats a reptile brought in from out of state as imported wildlife that needs a permit; **New Jersey's** permit list covers geckos as a family; and **Maine** clears geckos for captive-bred animals only. Your state wildlife agency is the answer that counts, and a city or a landlord can be stricter than the state.

Common mistakes & enrichment

The mistakes that cause the most harm

MISTAKE	WHAT IT CAUSES, AND THE FIX
Set up from a leopard gecko sheet at 30 to 40% humidity	Stuck shed on toes and tail tip. Run 50 to 70%, with a humid hide (page 8)
A humid hide that is allowed to dry out	The same. Check it daily; a dry humid hide is only a hide
Whole enclosure kept damp and cool	Respiratory infection. Moist hide, moderate air, warm side at target
A heat mat with no thermostat, or a heat rock	Belly burns. Mat on a thermostat, probe on the warm hide floor (page 7)
No calcium, no D3 and no UVB	Metabolic bone disease (page 24)
Pure sand, or a juvenile on loose substrate	Drying, shedding trouble, impaction risk. Soil mix for adults, paper for juveniles
Two geckos in one enclosure	Fights, and two males must never share. One per enclosure
No laying box for a female living alone	Egg binding risk. A box of moist substrate over the warm end (page 18)

Enrichment, by priority

No enrichment study has been run on this species. The closest evidence is a 2016 study of leopard geckos, the same family, by Bashaw and colleagues, which tested five types of enrichment: thermal and feeding enrichment produced the strongest responses, and mirror-based visual enrichment the weakest. Where fat-tails differ is humidity, which they need more of, and heat choice, where their own females pick hotter spots.

1. A properly maintained humid hide

A standing feature, never allowed to dry. Everything else on this list comes after it.

2. Thermal choice

Hides at both ends, so the gecko is hidden and at its chosen temperature at once, and a warm end that reaches the top of the range rather than sitting near it.

3. Feeding as hunting

Tong feeding, short supervised insect hunts in the enclosure, and varying when and where prey appears, all in the evening.

4. Floor space and digging

Cork flats, stacked slate and a deep patch of diggable substrate do more than any branch.

5. A laying box for any female

Husbandry and enrichment at once: a covered box she can dig in, over the warm end.

6. Novelty, last

A new piece of cork now and then. Which object a gecko takes to is individual. Skip mirrors, which ranked last, and never add a second gecko as enrichment.

Shedding, seasons, sounds & behavior

The shed cycle

Like a leopard gecko, a fat-tail goes through a complete shed cycle, in which the old outer skin loosens and comes off in pieces. In the days before a shed the colors go dull and the gecko hides more and may eat less. Fat-tails usually eat their shed skin, which recovers some of the nutrients in it, so an owner who never sees a shed is not missing anything. The day after, check the toes, the tail tip and both eyes.

SHED WEEK ROUTINE

Check the humid hide is damp and refresh it. No handling during the shed. If a meal is refused, skip it rather than pushing food. The day after, look at toes, tail tip and eyes in good light. Anything still attached gets the soak on page 23, never a fingernail.

Seasons

As daylight shortens in fall and winter, adults commonly eat less or stop for a period, and juveniles usually keep eating through it. That is normal in a healthy adult holding its weight. Some keepers, and one veterinary care sheet, run a cooler, drier winter cycle. A humid hide that never dries matters more than chasing an exact seasonal number, and this book uses one set of targets all year. Weigh every two weeks through any slowdown: a resting gecko lives off its tail and should not lose real weight.

Sounds and body language

YOU SEE OR HEAR	IT USUALLY MEANS
A chirp or squeak in the hand	Discomfort. Put it down
Quiet squeaks or clicks between males	A territorial dispute, which is why two males never share
Tail raised and waved slowly while watching an insect	Hunting. The same wave can also show nervousness
Arched back with a slow tail-swish at your hand	"Leave me alone"
Asleep in a hide all day, out at dusk	Normal for a nocturnal gecko
Digging and burrowing	Normal for this species. A female digging hard and restless may be about to lay

Normal or worth a look

NORMAL	WORTH A LOOK
Dull color and hiding for a few days, then a shed	Dull for over a week with no shed, or skin stuck on toes, tail tip or eyes
Eating less in fall and winter, weight steady	Eating less with a tail thinning toward the width of the neck
Sleeping by day in the warm hide	Lethargic in the evening, or hard to rouse at all
Using the humid hide often before a shed	Never leaving the humid hide, or never leaving the cool end: check temperatures

Health red flags & finding a vet

Fat-tails are hardy when their two biggest needs, heat and humidity, are met. Nearly every common problem traces back to one of them.

SEE A REPTILE VET PROMPTLY FOR

Wheezing, mucus at the nose or mouth, or open-mouth breathing. Lethargy, difficulty standing or walking, swollen joints, muscle twitches or a curving spine. Bloating, straining and appetite loss together. A female straining or digging with nothing produced. Retained shed tightening around a toe or the tail tip that home care has not cleared. Cloudiness, swelling or discharge in an eye outside a normal shed. Real weight loss, or a tail thinner than the neck. Tissue pushed out of the vent. Sunken eyes, diarrhea or an injury.

The husbandry connection

SETUP PROBLEM	WHAT IT CAUSES
Too dry, or no humid hide	Stuck shed on toes and tail tip, the most common complaint
Too cool and too damp together	Respiratory infection (page 25)
Calcium, D3 and UVB all missing	Metabolic bone disease (page 24)
Loose substrate in an unwell or cold gecko, oversized prey	Impaction
No laying site for a female	Egg binding (page 26)

What to tell the vet

- ☐ Warm hide floor reading from the infrared gun, not the thermostat setting
- ☐ Enclosure humidity, and whether the humid hide was damp
- ☐ Substrate, and the dusting routine: which powders, which days, UVB or not
- ☐ When it started and whether it is getting worse
- ☐ Last meal, last stool, and what the stool looked like
- ☐ The weight log, and a fresh stool sample in a sealed bag

Finding a vet before you need one

Not every practice treats reptiles. The ARAV (Association of Reptile and Amphibian Veterinarians) keeps a find-a-vet directory, and when you call a clinic, ask how many reptiles it sees. Book an **annual wellness exam** with a fecal test even when nothing is wrong: it covers body condition, hydration and nutrition, and the weight log is the most useful thing to bring. A routine exam commonly runs \$35 to \$105 and a fecal test \$65 to \$110; a sick visit with tests is typically \$150 to \$300 or more.

WHY WEIGHT COMES UP FIRST

A reptile hides illness until it is advanced, and a 50 g gecko hides weight loss until the tail has already thinned. A monthly number on a gram scale turns "seems thinner" into a fact a vet can use.

Retained shed & eye problems

The most common fat-tail-specific problem, and one of the most common reasons it sees a vet.

Retained shed (dysecdysis)

Cause	Humidity too low, or no humid hide, or one that has dried out. Less often parasites, poor nutrition, infection, or nothing rough to rub against. Dysecdysis means an abnormal or incomplete shed
Signs	Old skin left on the toes and the tail tip , and sometimes over the eyes. A ring of old skin left around a toe after the rest of the shed has come off
Response	Raise humidity with a properly damp humid hide. Soak the gecko for 15 to 30 minutes in shallow, warm water , never left alone, then ease the softened skin off a toe with a damp cotton swab. Repeat over a couple of days rather than forcing it. Never peel dry skin . See a vet if a ring has already tightened, the toe is swollen or darkened, or tissue is dying. A vet may soften it over hours and remove it under magnification
Recovery	Caught early, home care and a corrected hide usually clear it. Skin left to constrict cuts off circulation and can cost the tip of a toe or tail, and that loss is permanent

PREVENTION IS ONE BOX

A proper moist hide and correct humidity prevent almost all of this. Check toes, tail tip and eyes the day after every shed, and keep the moss damp all year, not only when a shed is due.

Eye problems

Cause	Reported relatively often in this species, and named by one exotic practice as its second most common problem. Retained eye caps from a bad shed are the usual cause. Vitamin A deficiency can swell the eyelids, and infection can follow either
Signs	Cloudiness, swelling or discharge outside a normal shed cycle, or an eye held closed
Response	A vet, promptly. Retained eye caps need veterinary attention rather than home removal. Never try to take anything off an eye yourself. Raise the humid hide's moisture in the meantime
Recovery	Follows the cause. The vet removes a retained cap safely, sometimes after softening it first. Fix the humidity, or it comes back with the next shed

Metabolic bone disease & vitamin problems

The most serious common problem in this species, and entirely a diet and setup disease.

MBD (metabolic bone disease)

Cause	A shortage of calcium or vitamin D3, often with no UVB. A poor calcium to phosphorus balance in the diet, or a warm side that runs too cool, adds to it. Veterinarians call it nutritional secondary hyperparathyroidism
Signs	Lethargy, difficulty standing or walking, swollen joints, muscle twitches, bones that feel unusually flexible, a curving spine. Later, a swollen or misshapen jaw, fractures from small falls, and loss of appetite
Diagnosis	A vet exam with x-rays, which show thin, poorly mineralized bone and any fractures, and blood tests
Treatment	The vet may give fluids, feeding support and calcium by injection if blood calcium is low. Correcting the diet and the setup is the core of treatment. The vet sets any dose; do not add extra D3 at home
Recovery	The outlook is favorable, especially if the gecko is still eating. Caught early it can be managed, but deformities already present are permanent, and advanced cases can be fatal

IN THE MEANTIME

Any tremor, any softness in the jaw or new trouble walking is a vet appointment this week. Handle the gecko as little as possible, and take out anything it can climb and fall from: weak bone breaks on a drop that would normally mean nothing.

Vitamin A deficiency

Cause	A diet short of preformed vitamin A, the ready-made form. Many reptiles cannot convert the plant form, beta carotene, so a multivitamin with preformed vitamin A is the prevention
Signs	Poor growth, poor appetite, swollen eyelids, mouth inflammation, and respiratory infections that follow
Diagnosis	From the diet history and the exam
Treatment and recovery	Diet correction and a supplement with preformed vitamin A, as the vet directs. Recovery follows the secondary problems, so the earlier the better

Gout

Cause	A buildup of uric acid that deposits in joints or organs. Too much protein, or dehydration and kidney trouble. A calcium powder that contains phosphorus can damage the kidneys and lead to it
Signs	Pain severe enough that the gecko will not move, eat or drink. Gritty or discolored urates are the early warning of the same uric acid chemistry
Diagnosis and treatment	X-rays show the deposits. The vet corrects the diet or the underlying problem and may prescribe long-term medication
Recovery	Typically poor once advanced. Prevention is water, the humid hide, and phosphorus-free calcium

Impaction & respiratory infection

One comes from what goes into the mouth, the other from the air the gecko sits in.

Impaction

Cause	A blockage of the gut. A ground gecko feeding on loose substrate can swallow some with its prey, and prey wider than the space between the eyes can lodge. It mostly happens in a gecko that is already unhealthy, dehydrated or kept too cool, because a cold gecko cannot move food through
Signs	Bloating, lethargy, appetite loss and straining , often with no normal stool
Diagnosis	A vet exam and x-rays, which show the blockage or swallowed material
Response	A vet visit, not a home remedy. Check the warm hide floor reads 88 to 92°F (31 to 33°C) the same day, and move the gecko onto paper towel
Recovery	Depends on what was blocking and how it had to be removed. Afterward, paper towel until the vet clears it, correctly sized prey from tongs, and a warm side checked weekly

PREVENTION

Paper towel for juveniles, quarantine and unwell geckos. A soil and sand mix rather than pure sand for adults. Prey no wider than the space between the eyes, offered from tongs or a dish. A warm side that is actually warm.

Respiratory infection

Cause	An enclosure kept too cool, too damp, or both . Poor hygiene, poor nutrition and vitamin A deficiency add to the risk
Signs	Wheezing, mucus, open-mouth breathing and lethargy . Discharge from the nose or eyes, and fast or labored breathing
Diagnosis	A vet exam, often with x-rays of the chest
Treatment	A vet and antibiotics. It will not clear on its own. The vet sets the drug and the dose. At home: clean conditions, the warm side at target, and the gecko kept toward the middle to upper end of its temperature range, which helps the immune system and thins the mucus
Recovery	Treatment works best started early, with the setup fixed at the same time. A severe or drawn-out infection can spread into the bloodstream

THE COMMON THREAD

Any gecko with either of these gets its warm hide floor, cool side and enclosure humidity measured the same day. The fix is usually in those three numbers, and the vet will ask for them.

Egg binding, prolapse & male problems

Reproductive emergencies. Every one of these is a vet visit, not a wait-and-see.

Egg binding (dystocia)

Cause	A female cannot pass eggs she has formed, with or without a male ever present. Causes include a missing or unsuitable nesting site, the wrong temperature or humidity, poor nutrition, weak muscles from too little exercise, oversized or misshapen eggs, an old pelvic injury, and illness
Signs	Restlessness, repeated digging in different spots, straining, a swollen vent, and a female who stops eating. As it goes on she turns dull and lethargic, and tissue may push out of the vent
Diagnosis	A vet exam with x-rays or ultrasound to find the eggs, and often blood work
Treatment	Ranges from a proper nesting site at the right conditions, through hormone injections or gently massaging the eggs out, to surgery. Medical treatment often fails, and surgery, which may remove the reproductive organs, is common. The vet decides the route and any doses
Recovery	Likely when she is in good overall condition and treated quickly. A female who has been egg-bound once can do it again, which is a conversation to have with the vet

NOT ALWAYS SUDDEN

In reptiles, egg binding is often not a sudden event. A female can hold eggs for weeks or months past the normal time to lay, which is exactly why a dull, digging, non-eating female is a same-day call rather than one to watch for a week.

Prolapse

Cause	An organ pushed out through the vent and stuck outside the body: the cloaca, the chamber just inside the vent, or the colon, the oviduct (the tube eggs pass down) in a female, or a hemipenis in a male. Parasites commonly cause it; so can a gut blockage, gut inflammation, bladder stones, straining with eggs, and injury
Signs	Pink, red or dark tissue outside the vent that does not go back in
Response	The same day. Do not push it back. Move the gecko onto clean, damp paper towel so the tissue does not dry out or pick up substrate
Diagnosis and recovery	The vet works out which organ it is and why it happened, and treats the cause as well as the tissue. The outlook depends on both, and on how quickly it was seen

Male problems

Impacted pores	The preanal pores of a male secrete a waxy scent. If they block, the area swells and may discharge. Signs: swelling or matter in the pore row in front of the vent. Response and recovery: a vet clears and treats them; do not squeeze them at home
Hemipenis prolapse	A male can prolapse a hemipenis. Treat it exactly as the prolapse above: damp paper towel, same-day vet

Parasites, mouth rot, tail loss & burns

Less dramatic than the last few pages. Each one still has a cause you can find and fix.

Internal parasites

Cause	Gut parasites, most of all in wild-caught animals, and spread between reptiles in a shared room or on shared tools
Signs	Weight loss, a tail thinning toward stick tail, repeated watery or foul-smelling stool, lethargy and appetite loss. A parasite load shows up in a fecal test long before it shows as weight loss
Diagnosis	A fecal exam: a fresh stool sample in a sealed bag, checked by the vet. Fecal exams at purchase and then on a regular schedule
Treatment and recovery	The vet treats when parasites are found and sets the drug and dose. Clean the enclosure thoroughly during treatment, and keep the gecko on paper towel so reinfection from droppings is easy to stop

Mouth rot (infectious stomatitis)

Cause	Infection and inflammation of the lining of the mouth, after an injury, with a nutritional deficiency, or in a poor environment
Signs	Tiny purplish red spots in the mouth are the early sign. Look whenever the gecko opens its mouth to eat
Treatment and recovery	The vet removes dead tissue, cleans the mouth with an antiseptic, and gives antibiotics and supportive care. Recovery follows how far it had spread, so the earliest spots are the time to go. Fix the underlying setup or diet problem at the same time

Tail loss

Cause	Rough handling, grabbing the tail, or a bad fright. A gecko that drops its tail again and again may have an underlying health problem
Signs	The tail is off. The gecko is otherwise usually fine
Response	Move it onto clean paper towel, keep the enclosure clean while the stump heals, and skip handling. A stump that swells, oozes or smells is a vet visit
Recovery	The tail regrows within weeks, smoother and more bulbous than the original. The real loss is the fat reserve it held, so watch the weight closely for the next few months

Burns

Cause	A heat source the gecko can reach or one that is not regulated: a heat rock, an unscreened bulb, a mat with no thermostat or a failed one
Signs	Damaged, discolored or blistered skin, usually on the belly, and a gecko that may lose fluid and become dehydrated
Response	Turn the heat source off, find out why the regulation failed, and see the vet. The vet cleans the burn, treats it against infection, and gives pain relief and fluids
Recovery	Many burns respond well to treatment. Severe ones have a guarded outlook. Keep the gecko on clean, dry paper towel while it heals

Reading poop & hydration

Stool and urates are a free daily health check. Look before you scoop.

A reptile passes nitrogen waste as **urates**: a white to cream, chalky or paste-like portion that comes out alongside the solid stool, a reptile's version of urine. The urate is the fastest daily read on hydration a keeper has. Adult geckos usually pass stool every few days to about weekly, and juveniles more often; learn your own gecko's pattern.

IT LOOKS LIKE	USUALLY MEANS	DO
A firm, defined log with a soft white urate	Normal. Small bits of insect shell in it are normal too	Remove it, note it in the log
Watery or unformed for more than one stool in a row	Often parasites, sometimes a feeder change or stress	Fresh sample in a sealed bag, fecal test
Blood, bright red or dark and tarry	Bleeding somewhere in the gut	Vet
Straining, or nothing passed with a swollen belly	Impaction, eggs, or a vent problem	Check the warm side, then vet (pages 25 and 26)
An unusually strong or off smell	A gut problem, often parasites	Fecal test
Urate yellow, orange, or gritty and crystalline	Dehydration at least; uncorrected, a urate or kidney problem, the chemistry behind gout	Soak, refresh the humid hide and the water. If it persists, vet (page 24)

Any of these with lethargy or a skipped meal is worth a call that week, not a wait.

Hydration

- **Shallow water dish, always**, emptied, rinsed and refilled every day.
- **The humid hide is hydration equipment**, not only shed equipment. Keep the moss damp all year.
- **A light misting** a few times a week in a dry house gives droplets as well as raising the air.

SIGNS OF DEHYDRATION

Loose or wrinkled skin with no shed due, sunken eyes, hard or yellow urates, stuck shed, and a drop in activity and appetite. A mild case: a **15 to 30 minute soak in shallow, warm water**, never left alone, then a properly damp humid hide and fresh water. A gecko that stays wrinkled or sunken-eyed after that needs a vet, because dehydration that persists has something behind it, and shedding-related dehydration that runs past about a week can become fatal.

BRING A SAMPLE

Take a fresh stool in a sealed bag to every wellness exam, even when nothing looks wrong. A baseline fecal turns a first vet visit into a real record, and parasite trouble shows there before it shows on the gecko.

Setup checklist & targets

Essential setup checklist

- ☐ 36×18×18 in enclosure minimum, or a 40-gallon breeder, with a mesh top or front opening
- ☐ Under-tank heat mat, about a third of the floor
- ☐ Thermostat, probe on the warm hide floor
- ☐ Infrared temperature gun, digital thermometer and hygrometer
- ☐ Three hides: warm, cool and humid
- ☐ Sphagnum moss for the humid hide
- ☐ Topsoil and play sand for 4 in (10 cm), or paper towel for a juvenile
- ☐ Shallow water dish
- ☐ Digital gram scale
- ☐ Plain phosphorus-free calcium, calcium with D3, and a multivitamin
- ☐ Feeding tongs and a gut-load diet for the insects
- ☐ Light timer, plus optional low-output 2 to 5% UVB
- ☐ Cork flats and slate, set on the tank floor
- ☐ For a female: a covered laying box

Targets

TARGET	°F	°C
Warm side, surface of the warm hide floor	88 to 92	31 to 33
Cool side	75 to 80	24 to 27
Night	70 to 75	21 to 24
Act if the night falls below	70	21
Humidity	50 to 70% in the enclosure, 70 to 80% inside the humid hide	
Light	10 to 12 hours a day on a timer, lights off at night	
Adult weight	40 to 90 g, about 50 g on average, weighed monthly	

Before your gecko comes home

- ☐ Enclosure running for a couple of days, warm hide floor verified at 88 to 92°F (31 to 33°C)
- ☐ Humid hide filled and damp; hygrometer reading 50 to 70%
- ☐ A week of insects on gut-load, and both calcium powders on hand
- ☐ Reptile vet found, first exam booked, quarantine space ready (page 19)

Emergency & quick targets

CALL A REPTILE VET NOW IF

A female is straining to lay, or digging and dull and off food. There is wheezing, mucus, or open-mouth breathing. The belly is bloated with straining and no appetite. The gecko has trouble standing or walking, swollen joints, twitching muscles or a curving spine. Tissue is pushed out of the vent. The belly skin is burned. The gecko is limp or cannot be roused.

BOOK THIS WEEK IF

Retained shed is tightening around a toe or the tail tip after home care. An eye is cloudy, swollen or weeping outside a shed. The tail has thinned below the width of the neck, or the weight is sliding. Urates stay yellow or gritty after a soak.

READING	°F	°C
Warm hide floor	88 to 92	31 to 33
Cool side	75 to 80	24 to 27
Night, and the point to act	70 to 75; act below 70	21 to 24; act below 21
Humidity	50 to 70% enclosure, 70 to 80% in the humid hide	

Your numbers

Reptile vet _____

Normal weight (g) _____

24-hour exotic emergency clinic _____

UVB tube installed _____

Gecko's name, sex, hatch date _____

Last vet visit _____

Routine, written down

Feeding days _____

Calcium with D3 days _____

Plain calcium days _____

Multivitamin day _____

HAVE THIS READY WHEN YOU CALL

Weight in grams and the last three from the log. The warm hide floor reading from the infrared gun in the last hour, and the humidity. Substrate type. Dusting routine and whether UVB is running. When it last ate and last passed stool. A photo of anything visible.

Budget & shopping list

United States retail ranges, in dollars. The low end is basic but correct gear; the high end is premium.

INITIAL SETUP	TYPICAL COST
Enclosure, 36×18×18 in (a 40-gallon breeder gives the same floor)	\$202 to \$258
Under-tank heat mat with a quality thermostat	\$40 to \$70
Three hides: warm, cool and humid	\$20 to \$35
Sphagnum moss for the humid hide	\$5 to \$10
Substrate: coconut fiber and reptile sand for the burrowing mix	\$19 to \$31
Digital thermometer and hygrometer, infrared gun ideal	\$15 to \$40
Digital gram scale	\$15 to \$40
Shallow water dish	\$8 to \$15
Low-output UVB, optional but beneficial	\$45 to \$80
Equipment subtotal	\$369 to \$579
First vet exam (\$35 to \$105) and fecal test (\$65 to \$110)	\$100 to \$215
All-in before the gecko	\$469 to \$794

The gecko is separate: \$75 to \$600, with normal, wild-type patterned animals commonly around \$100 and usually \$100 to \$200. Mid-range morphs such as amelanistic (lacking dark pigment) or single-gene Oreo run \$200 to \$400; Whiteout, Zulu and stacked multi-gene animals can reach \$600 to \$1,000 or more. Without UVB, the equipment subtotal is \$208 to \$420.

EVERY YEAR	TYPICAL COST
Feeder insects and supplements, at \$10 to \$25 a month	\$120 to \$300
Annual wellness exam and fecal test	\$100 to \$215
A healthy adult, per year	\$220 to \$515

On top: electricity for the mat and any UVB, which is modest; a replacement UVB tube about every 12 months if you run one; fresh substrate at each monthly change; and a laying box for a female, which costs little next to an egg-binding surgery. Over 15 years the yearly costs come to about \$3,300 to \$7,700.

PLAN FOR, DON'T COUNT ON AVOIDING

An illness or emergency visit with tests typically runs \$150 to \$300 or more, and surgery, for egg binding for example, costs more still. A few hundred dollars set aside matters as much as any line above.

WHERE NOT TO CUT CORNERS

The thermostat, the humid hide and the scale. Each costs little, and each prevents one of the three problems this gecko runs into most: burns, stuck shed, and weight loss nobody noticed.

First 30 days

Day 1

- ☐ Verify the warm hide floor with the infrared gun before the gecko goes in
- ☐ Weigh it in grams and photograph it and the setup, for a baseline
- ☐ Paper towel substrate if it is a juvenile or in quarantine
- ☐ Fill in page 30 and the equipment dates on page 38

Week 1

- ☐ No handling. Fresh water daily, humid hide checked daily
- ☐ Offer food in the evening on its age schedule, and remove what is not eaten
- ☐ Hiding and refusing food are normal after a move. Note stool and urates
- ☐ Book the first vet exam, and take a fresh stool sample

Week 2

- ☐ Confirm it uses the warm hide and the humid hide. If it never does, re-check temperatures and hide size
- ☐ Once it is eating regularly, about two weeks in, start short evening handling sessions, low over a bed or the floor

Weeks 3 to 4

- ☐ Watch the first shed, then check toes, tail tip and eyes the day after
- ☐ Weigh again and start the Owner Log on page 37
- ☐ Set the dusting days and write them on page 30
- ☐ Settle the feeding interval on the tail, not the calendar (page 12)

Days 31 to 90 and beyond

- ☐ Act on the fecal result rather than filing it, and book any follow-up
- ☐ Joining an existing collection? Hold the full 3 to 6 month quarantine before it shares a room, tools or hands (page 19)
- ☐ A healthy adult out of quarantine can move from paper to the 4 in (10 cm) soil mix
- ☐ A female: add the laying box now, before she needs it (page 18)
- ☐ Fill in the pet-sitter sheet on page 36 before you need it
- ☐ Look back over the log for any slow drift in weight, appetite or tail

THE FIRST MONTH IN ONE LINE

Leave it alone, keep it warm on one side and damp in one box, feed it in the evening, and weigh it. Almost everything that goes wrong early shows up as a warm side that is not warm, a humid hide that is not damp, or a weight that is not holding.

Symptom quick reference

Find the sign, read the likely cause, then turn to the page. When two rows fit, the vet decides between them.

SYMPTOM	LIKELY CAUSE	ACTION
Skin stuck on toes or the tail tip	Retained shed: humidity too low, no humid hide	Damp hide, warm soak, swab; vet if a ring has tightened (page 23)
Cloudy, swollen or weeping eye outside a shed	Retained eye cap, vitamin A deficiency, infection	Vet, do not touch the eye (page 23)
Trouble walking, twitching, swollen joints, bent spine	Metabolic bone disease	Vet this week, remove anything climbable (page 24)
Swollen eyelids, poor appetite, mouth sores	Vitamin A deficiency	Vet; check the multivitamin (page 24)
Will not move, eat or drink; gritty urates	Gout	Vet (page 24)
Bloated belly, straining, no appetite	Impaction	Check the warm side, paper towel, vet (page 25)
Wheezing, mucus, open-mouth breathing	Respiratory infection	Vet; warm side to target, less damp (page 25)
Female digging, restless, straining, off food	About to lay, or egg binding	Laying box now; vet the same day if straining or dull (page 26)
Tissue outside the vent	Prolapse	Damp paper towel, vet the same day (page 26)
Swelling or matter in a male's pore row	Impacted preanal pores	Vet (page 26)
Tail thinning, watery or foul stool	Internal parasites	Vet with a fresh sealed sample (page 27)
Purplish red spots in the mouth	Mouth rot	Vet (page 27)
Damaged or blistered belly skin	Burn	Heat off, fix the thermostat, vet (page 27)
Tail dropped	Grabbed or frightened	Paper towel, keep clean, no handling (page 27)
Wrinkled skin, sunken eyes, yellow urates	Dehydration	Warm soak, damp hide, fresh water; vet if it persists (page 28)
Off food, otherwise bright, tail full	Season, shed, stress, or a cool warm side	Measure the warm hide floor first (page 15)
Weight sliding between log entries	Several possible causes	Vet, bring the Owner Log (page 37)

Daily, weekly & seasonal routine

Daily

- ☐ Empty, rinse and refill the water dish
- ☐ Squeeze the humid hide moss: damp, not dripping, not dry
- ☐ Read the second thermometer, apart from the thermostat probe
- ☐ Remove droppings and uneaten insects; glance at stool and urates
- ☐ See the gecko at dusk. One that is not out in its own activity window is the earliest soft signal there is

Feeding days

- ☐ Evening feeding on the age schedule, from tongs, dusted per the day
- ☐ Next batch of insects on gut-load 24 to 48 hours ahead

Weekly

- ☐ Infrared gun on the warm hide floor; hygrometer reading logged
- ☐ Wipe down hides, dish and glass
- ☐ Check the thermostat probe is still taped in place
- ☐ Look at the tail and fill in a row of the Owner Log

Monthly

- ☐ Weigh in grams and log it (page 37)
- ☐ Deep clean and replace the substrate; replace the moss
- ☐ Review the log for slow trends, not single readings

Seasonal

- ☐ Fall: expect appetite to drop in an adult. Weigh every two weeks through it
- ☐ Winter: check the night low in the room. Below 70°F (21°C), add heat with no light
- ☐ Winter: heating dries the air. Watch the hygrometer and the humid hide
- ☐ Summer: check the warm side on the hottest afternoon. The thermostat should be cutting out
- ☐ Breeding season, for a female: laying box damp and in place

Yearly

- ☐ Wellness exam and fecal test
- ☐ Replace the UVB tube, even if it still lights
- ☐ Replace opened supplement powders
- ☐ Re-read pages 29 and 30 against the real setup. Things drift

IF YOU ONLY KEEP TWO HABITS

The daily squeeze of the humid hide and the monthly weigh-in. One prevents this gecko's most common problem, and the other catches most of the rest before they show.

Power outages, travel & transport

A fat-tail's normal night low is 70 to 75°F (21 to 24°C). Act once it is heading below 70°F (21°C). A small gecko loses heat faster than a heavy-bodied snake in the same room.

When the power goes out

Do

- Put a battery or wind-up thermometer in the enclosure so you know the real number.
- Drape thick towels or a blanket over the enclosure, leaving a vent gap, or move the gecko in a ventilated tub to the warmest room, usually one without outside walls.
- Set a wrapped hot water bottle or a commercial hand warmer beside the gecko, never touching it.
- Stop feeding insects until heat is reliably back. Keep water available and the humid hide damp.
- Past a day or two below the floor, board it with a vet, an experienced friend with power, or a reptile shop.

Don't

- No candles, gas stoves or open flame near the enclosure: a fire and carbon monoxide risk.
- No heat pack against the skin. A cold, sluggish gecko cannot move off it and will burn.
- No food "to keep its strength up". Undigested food in a cold gut is a risk, not a benefit.
- No hot water to warm a cold, unresponsive gecko.

OUTAGE KIT

Two or three hand warmers or a hot water bottle, two thick towels, a battery thermometer, a ventilated tub, and both vet numbers written on paper.

Leaving town

TRIP LENGTH	ADULT	JUVENILE, OR A GECKO THAT IS SICK, SHEDDING OR GRAVID
1 to 2 nights	Often fine alone. Timers and thermostat running, fresh water and a damp humid hide before you go	A sitter once a day for food, water, the hide and a temperature check
3 to 7 nights	A sitter every 1 to 2 days for water, the humid hide, a stool check and a temperature glance. Feed on the normal schedule	A sitter daily, with pre-portioned insects and written dusting days
Over a week	A daily sitter who has done a practice visit with you, or boarding with a reptile-experienced vet or shop	

Transport and vet trips

- A ventilated plastic tub with a towel to grip beats a wire carrier: it holds heat. Carry it in a dark bag or box. No loose substrate, no open water dish.
- Skip the meal before travel. Warm the car first, and for any trip over about 30 minutes below 65°F (18°C), put a wrapped hand warmer beside the tub, never against the gecko.
- Check the tub every 20 to 30 minutes at a stop. Never leave the gecko in a parked car, hot or cold, even briefly.
- Moving house: set up the new enclosure and check its temperatures before the gecko arrives.

FILL IN AND LEAVE BY THE ENCLOSURE

Pet-sitter sheet

Everything a sitter needs on one page. Walk them through it once in person before the first solo visit.

Gecko's name _____

Reptile vet _____

Normal weight (grams) _____

24-hour emergency clinic _____

My phone _____

Backup contact _____

The numbers

READING	SHOULD BE
Warm hide floor (point the infrared gun at it)	88 to 92°F (31 to 33°C)
Cool side	75 to 80°F (24 to 27°C)
Humidity meter	50 to 70%
Lights	On at _____ and off at _____, on a timer

Each visit

- | | |
|---|--|
| ☐ Fresh water in the shallow dish | ☐ Remove droppings and note whether they looked normal |
| ☐ Squeeze the moss in the humid hide, and dampen it if it is dry | ☐ Feed on the days below, in the evening, and remove what is left |
| ☐ Check the two temperatures above | ☐ Wash hands with soap after. Nothing goes in the kitchen sink |

Feeding while I'm away

Feed on these days _____

Insects are kept _____

How many insects _____

Dust with (which tub, which days) _____

Please don't

- Pick the gecko up by the tail, ever. It will drop it. Better still, do not take it out at all.
- Change any dial, timer or thermostat, or move the lights or the probe.
- Feed anything not on this sheet, or catch insects outside.

CALL ME IF

The warm hide reads well outside 88 to 92°F (31 to 33°C), the room is below 70°F (21°C) at night, or the lights are off in the daytime. The gecko is breathing with its mouth open, wheezing, or is limp and hard to rouse. Anything is sticking out of the vent. The tail has come off. A female is straining. No movement across two visits. If I don't answer, call the vet.

Owner log

[illegible]

A weight trend is the best early warning you have, and with this gecko it is doubly true: the tail hides a slow loss until it is well along, and a healthy seasonal slowdown and an illness look the same from outside. The scale and the humid hide column tell them apart. Sixteen rows is four months, long enough to see a real trend.

Equipment, supplement & vet log

Equipment log

ITEM	INSTALLED OR OPENED	REPLACE OR CHECK BY	DONE
UVB tube, if used (about every 12 months)			
Heat mat and thermostat (verify monthly)			
Substrate (replace at the monthly deep clean)			
Thermometer, hygrometer and gun batteries			
Calcium, calcium with D3, multivitamin (opened)			

Twelve-month planner

Tick the box in the month you did it. Start with whichever month you are in.

TASK, BY MONTH	1	2	3	4	5	6	7	8	9	10	11	12
Weigh and log	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Deep clean, new substrate	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Verify warm hide floor	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Wellness exam, fecal	☐											☐

Dusting tally

For a month or two, mark each feeding with C (plain calcium), D (calcium with D3) or M (multivitamin), so the weekly pattern can be checked against page 14.

WEEK OF	MON	TUE	WED	THU	FRI	SAT	SUN

Vet visit log

DATE	GRAMS	REASON AND FINDINGS	TREATMENT, FOLLOW-UP

REFERENCE

Glossary, A to I

Every abbreviation and technical word used in this package, in plain words.

Amelanistic	A morph lacking dark pigment, so it looks paler than a normal fat-tail.
ARAV	Association of Reptile and Amphibian Veterinarians, which keeps a find-a-vet directory.
Autotomy	Dropping the tail on purpose when grabbed or frightened. The tail regrows within weeks, smoother and more bulbous.
Beta carotene	The plant form of vitamin A, which many reptiles cannot convert. Hence a multivitamin with preformed vitamin A.
Black soldier fly larvae	A feeder grub that, unlike most feeders, carries close to as much calcium as phosphorus.
Calcium to phosphorus ratio	How much calcium a food holds against phosphorus. About 2 to 1 is the target; feeder insects run far below it.
Captive-bred	Hatched and raised in captivity. The opposite of wild-caught, and the one to buy.
CDC	Centers for Disease Control and Prevention, the United States public health agency that tracks <i>Salmonella</i> outbreaks.
Ceramic heat emitter	A screw-in heater that gives heat and no light, for cold nights. Always on its own thermostat.
Cloaca	The chamber just inside the vent that waste, urates and eggs all pass through.
Clutch	One laying of eggs: 1 to 3 in this species.
Crepuscular	Most active at dusk and dawn.
Dusting	Coating feeder insects in supplement powder just before they are offered.
Dysecdysis	An abnormal or incomplete shed: retained shed. Page 23.
Dystocia	Egg binding: a female unable to pass eggs she has formed. A same-day vet problem. Page 26.
Eublepharid	A member of the eyelid gecko family, Eublepharidae: the fat-tail, the leopard gecko and their relatives.
Fecal test	A vet's check of a fresh stool sample for parasites.
Ferguson Zone	One of four groups that reptiles are sorted into by how much ultraviolet light they seek in the wild. The fat-tail is Zone 1, the shade dwellers.
Gout	A painful buildup of uric acid in the joints or organs. Page 24.
Gravid	Carrying eggs.
Gut-loading	Feeding feeder insects a calcium-rich diet for 24 to 48 hours before they are offered, so the gecko eats it too. Page 14.
Hemipenes	The paired male reproductive organs, which sit in the base of the tail and make it look more bulbous.
Humid hide	A closed hide of damp moss at 70 to 80% humidity inside, in an enclosure that runs 50 to 70%. Page 8.
Hygrometer	A humidity meter. Digital ones are worth the small extra cost.
Impaction	A blockage of the gut, from swallowed substrate or prey too large to pass. Page 25.
Infrared temperature gun	A handheld thermometer that reads the temperature of a surface from a distance. The way to check the warm hide floor.
IUCN	International Union for Conservation of Nature, which keeps the Red List of species at risk. The fat-tail is listed as Least Concern.

REFERENCE

Glossary, M to Z

MBD	Metabolic bone disease: weak, poorly mineralized bone from too little calcium or vitamin D3. Page 24.
Mites	Tiny parasites that live on a reptile's skin. Paper towel during quarantine shows them.
Morph	A captive-bred color or pattern line, such as amelanistic, Oreo, Whiteout or Zulu.
Nocturnal	Active at night, asleep by day.
Obligate insectivore	An animal that must eat insects and cannot use plant food. That is the fat-tail.
Oxalic acid	A compound in spinach, beet greens and chard, also called oxalates, that binds calcium so it cannot be absorbed.
Oviduct	The tube a female's eggs pass down before they are laid.
Preanal pores	The chevron-shaped row of small openings in front of the vent, enlarged in males, which produce a waxy scent. Page 17.
Preformed vitamin A	Vitamin A in its ready-made animal form, which reptiles can use directly.
Probe	The thermostat's sensor. It goes flat on the warm hide floor. Page 7.
Prolapse	An organ pushed out through the vent and stuck outside. Same-day vet. Page 26.
Pulse proportional thermostat	A controller that sends short bursts of power to hold a steady temperature. A dimming one turns power down like a light dimmer. Both suit fast heat sources; an on/off unit suits a mat.
Quarantine	Keeping a new animal apart, in another room, for 3 to 6 months before it shares air, tools or hands with others. Page 19.
<i>Salmonella</i>	Bacteria that healthy reptiles carry in their droppings and that can make people ill. Hand-washing is the control. Page 16.
Stick tail	A tail wasted down to skin over bone: an underweight gecko that needs a vet and a parasite check.
Stomatitis	Mouth rot: infection and inflammation of the lining of the mouth. Page 27.
Substrate	Whatever covers the enclosure floor: paper towel, or a soil and sand mix.
Temperature-dependent sex determination	When the egg's incubation temperature, not genes, decides whether the hatchling is male or female. Page 17.
Thermal gradient	The warm-to-cool range from one end of the enclosure to the other, which lets the gecko choose its temperature.
Thermostat	The controller that switches the heat source to hold a set temperature. The device that prevents burns.
Urates	The white to cream, chalky part of a dropping: uric acid, a reptile's version of urine. Yellow or gritty means dehydrated. Page 28.
Uric acid	The form most reptiles pass nitrogen waste in. When it builds up in the body, it is gout.
UVB	Ultraviolet B, the band of ultraviolet light a reptile uses to make vitamin D3 in its skin. Glass blocks most of it. Page 10.
UVI	Ultraviolet index, the measured strength of UVB where the gecko sits. Zone 1 target: 0 to 0.7 across the enclosure, 0.6 to 1.4 at the closest point.
Vent	The single rear opening that waste, urates and eggs pass through.
Vitamin D3	The vitamin that lets a reptile absorb and use calcium. Made in the skin under UVB, or taken in from calcium with D3 powder. Builds up if overdosed.
Wild-caught	Taken from the wild rather than bred. Higher parasite risk and fecal tests matter most. Buy captive-bred.

REFERENCE

Sources

Where the numbers in this package come from, so you can check them yourself.

Beastly Facts guides, the primary source

Every husbandry figure that the Beastly Facts site carries is printed here exactly as the site states it: the African fat-tailed gecko enclosure setup, feeding, health issues, handling, enrichment and cost guides, the fat-tail encyclopedia entry and care guide, and the cross-species reptile guides on heating and thermostats, quarantine, *Salmonella* and hygiene, stool and urates, shedding, gut-loading, UVB lighting, and the reptile emergency plan, which carries the fat-tail's night floor. The leopard gecko figures on page 5 come from the site's leopard gecko guides. State rules come from the Beastly Facts state legal research.

Veterinary and species references

- Tree of Life Exotic Pet Medical Center: African Fat-Tailed Gecko Care. The temperature and 50 to 70% humidity figures, and the older 20-gallon enclosure floor; gut-loading 24 to 48 hours ahead; the 10 to 20 year veterinary lifespan figure; a multivitamin once a week; fecal exams for new and wild-caught geckos; annual wellness visits.
- Cadillac Veterinary Clinic: African Fat-Tailed Geckos. Untreated topsoil in the 70 to 30 mix and keeping it moisture-maintained; decorations that bear the gecko's weight; eggs every 15 to 22 days in a 4 to 5 month season and sperm storage; impacted pores and hemipenis prolapse in males; eye problems as the second most common presentation; stick tail and the parasite check; mouth rot; the vet triggers used on page 19.
- Animal Diversity Web, University of Michigan: *Hemitheconyx caudicinctus*. Range, weight and tail thickness, clutch size and number, incubation length, temperature-dependent sex determination, maturity, eating the shed, tail-waving.
- Reptiles Magazine, Craig Stewart: African Fat-Tailed Gecko Care Sheet. Hatchling size, the hatchling and juvenile cricket counts, a vitamin powder about once a week, misting several times a week.
- ReptiFiles: African Fat-Tailed Gecko Care Sheet. The 36×18×18 in minimum enclosure for one adult.
- Bragg, Fawcett, Bragg and Viets (2000), Biological Journal of the Linnean Society: nest-site selection in two eublepharid geckos. The 32.4°C and 28.7°C nest choices.
- Merck Veterinary Manual: Nutritional, Metabolic and Endocrine Diseases of Reptiles. Metabolic bone disease causes, signs, x-ray and blood diagnosis, treatment and outlook; vitamin A deficiency; gout; obesity.
- Merck Veterinary Manual: Environmental Diseases and Traumatic Injuries of Reptiles. Burns, and the causes of retained shed.
- Merck Veterinary Manual: Disorders and Diseases of Reptiles. Respiratory infection causes, signs and the warmer-range rule; mouth rot; prolapse; egg binding that is not sudden and often needs surgery; dehydration signs; fecal testing.
- PetMD: Egg Binding in Reptiles. Causes, signs, diagnosis, treatment options and recovery; females laying without a male.
- Centers for Disease Control and Prevention: *Salmonella* outbreak linked to pet geckos, closed January 2026. Cases, hospitalizations, states, and the under-5 rule.

The 2016 leopard gecko enrichment study by Bashaw and colleagues on page 20 is reported as the Beastly Facts enrichment guide reports it. The calcium to phosphorus figures for feeder insects on pages 13 and 14 come from the Beastly Facts gut-loading guide, which credits Mark Finke's analyses. The 3 to 6 month quarantine comes from the Beastly Facts quarantine guide, which cites the Merck Veterinary Manual.

REFERENCE

Where the sources disagree, version history & about

Where credible sources give different numbers, this package prints one and says why.

TOPIC	WHAT THE SOURCES SAY	WHAT THIS PACKAGE USES
Humidity	50 to 70% (Beastly Facts, Tree of Life); about 30 to 40% with light misting (Cadillac Veterinary Clinic); "slightly more humid than leopard geckos" (Reptiles Magazine)	50 to 70%, plus a humid hide at 70 to 80%. The lower figure is the leopard gecko range, and dry enclosures are what cause stuck shed
Habitat	Savanna, dry forest and rocky scrub; arid and semi-humid grassland; one care sheet says "desert areas"	Savanna and scrub. The desert description is the outlier
Lifespan	15 to 20 years commonly cited; best-documented captive record just over 16; 10 to 20 in two veterinary and academic references	15 to 20 commonly cited, the record just over 16. Plan on well over a decade
Adult feeding	3 times a week; 3 to 4 times a week; every 2 to 3 days for young adults; every 5 days for a fat-tailed mature adult	The full range, corrected by the tail
Plain calcium	Every feeding; at least 3 meals a week	Most feedings, inside that range
Multivitamin	Once a week (two references); not stated on the site	About once a week
Gut-load timing	24 to 48 hours (species guide, Tree of Life); 24 to 72 hours (gut-loading guide)	24 to 48 hours, inside the 24 to 72 hour window
Enclosure	A 10 gallon for a pair (one care sheet); 20-gallon long for one adult (veterinary care sheets); 36×18×18 in for one adult (a specialist care sheet)	36×18×18 in minimum, one gecko, the same floor as the leopard gecko
UVB	Not required; required; 2 to 5% increasingly recommended	Optional with D3 powder, 2 to 5% recommended

Version history

VERSION	TEMPLATE	WHAT CHANGED
1.0 · Oct 2026	t3	First edition, 42 pages. Built new on the t3 template from the Beastly Facts African fat-tailed gecko guides, the shared reptile guides, and the veterinary references listed under Sources.

This package is a husbandry reference compiled from the listed sources and cross-checked against current veterinary guidance at the time of publication. It is not a substitute for a veterinarian who sees reptiles, and nothing in it should delay a vet visit. Treatments described here are for recognizing what a vet may do, not for use at home, and every dose is the vet's to set.

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